INCH-POUND

MIL-B-44059D
3 September 1993
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MILITARY SPECIFICATION

BEEF STEW, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers thermostabilized beef stew in flexible pouches for use by the Department of Defense as a component of the Meal, Ready-To-Eat, Individual.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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SPECIFICATION

MILITARY

MIL-P-44073 - Packaging and Thermoprocessing of Foods in Flexible Pouches

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks, and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Carrots for Processing (7 CFR 51.4140)
U.S. Standards for Grades of Fresh Shelled Peas for Canning or Freezing (7 CFR 51.4210)
U.S. Standards for Grades of Tomato Paste (7 CFR 52.5041)

(Copies are available from the Director, Processed Products Branch, Fruit and Vegetable Division, Agricultural Marketing Service (AMS), U.S. Department of Agriculture, Room 2077, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

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Institutional Meat Purchase Specifications for Fresh Beef, Series 100

(Copies are available from the Director, Livestock and Seed Division, Agricultural Marketing Service (AMS), U.S. Department of Agriculture, Room 2603, South Building, P.O. Box 96456, Washington, DC 20090-6456.)

Meat and Poultry Inspection Regulations (9 CFR Parts 301-399)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)
U.S. Standards of Identity for Margarine

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, One Maritime Plaza, 23rd Floor, San Francisco, CA 94111.)

AOAC INTERNATIONAL

Official Methods of Analysis of the AOAC

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(Application for copies should be addressed to AOAC International, 2200 Wilson Boulevard, Suite 400, Arlington, VA 22201-3301.)

NATIONAL ACADEMY OF SCIENCE

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, NW, Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent or insect infestation, extraneous material, off-odors, off-flavors, and off-colors.

3.2.1 Beef. The beef shall be from steers, heifers, or cows and shall be derived from any one or any combination of the following USDA Certified Institutional Meat Purchase Specifications (IMPs) cuts: Item No. 114 (chuck, shoulder clod), Item No. 161 (round, shank-off, boneless), Item No. 163 (round, shank-off, 3-way, boneless), Item No. 165 (round, rump, and shank-off, boneless), Item No. 167A (round, knuckle, peeled), Item No. 169 (round, top, (inside)), Item No. 170A (round, bottom, (gooseneck), heel out), and Item No. 171B (round, outside round). All beef shall be certified by a USDA, Agricultural Marketing Service (AMS) agent for condition of the product and compliance with the Institutional Meat Purchase Specifications Quality Assurance Provisions.

NOTE: Certified IMPS cuts may be further trimmed (see 3.3.1) and broken down into smaller cuts after certification, as requested by the purchaser. Boxes used for shipment of the further trimmed and broken down cuts shall be labeled as "Beef for further processing as specified in MIL-B-44059". The certificate shall be noted as such and shall accompany the shipment.

3.2.1.1 Handling and storage. Handling and storage of the boned and trimmed beef, prior to processing into the finished product, shall be in accordance with the following requirements:

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a. Beef processed on the day of initial certification shall be maintained in the temperature range of 28°F to 50°F, inclusive.

b. Holding in the fresh-chilled state for not more than four days after initial certification is permitted, provided that the beef is maintained in the temperature range of 28°F to 40°F, inclusive.

c. Holding in the frozen state for not more than 180 days after initial certification is permitted, provided that the beef is:

- frozen to 0°F or lower within 72 hours after initial certification
- stored at 0°F or lower
- protected from freezer deterioration and damage
- stored in containers that are adequate to maintain product excellence

d. Tempering/thawing of frozen beef shall comply with the following:

- microwave tempering/thawing of frozen beef is permitted provided excellent condition of the beef, as determined by USDA, is maintained
- the microwave tempering/thawing process shall not exceed 48 hours from time of removal of the beef from the freezer to further preparation
- regardless of the process used, frozen beef shall be tempered/thawed to a temperature not to exceed 40°F
- when a tempering/thawing process other than microwave is used, the time from the start of the tempering/thawing process (removal from freezer) to further preparation shall not exceed 96 hours
- beef which has been tempered/thawed shall not be refrozen.

3.2.1.2 Beef, cooked, frozen. The cooked frozen beef logs or dices shall be prepared from beef that complies with the requirements of 3.2.1 and 3.2.1.1. The beef shall be processed in accordance with 3.3.1, 3.3.1.1, and 3.3.1.2, as applicable. The cooked beef logs or dices shall be packaged and sealed in a USDA approved food packaging material and shall be frozen to an internal temperature of 0°F or below within 72 hours. The initial freezing (in-storage) date, and processing and packaging requirements shall be certified by a USDA Agricultural Marketing Service (AMS) certificate. The packaged, frozen, cooked beef logs or dices shall be held at an internal temperature of 0°F or below for a period not to exceed 120 days prior to product preparation (see 3.4).

3.2.2 Potatoes.

3.2.2.1 Potatoes, fresh. The potatoes shall be fresh, firm, clean, sound, and of a white flesh variety suitable for canning. The maximum specific gravity for the potatoes shall be 1.075 with a reducing sugar content of not more than 2 percent on a dry weight basis.

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3.2.2.2 Potatoes, diced, dehydrofrozen,. The diced, dehydrofrozen potatoes shall be prepared from fresh potatoes meeting the requirements of 3.2.2.1. The dehydrofrozen potatoes shall be of the latest season's pack and shall be mechanically diced as specified in 3.3.2.1.

3.2.3 Carrots.

3.2.3.1 Carrots, fresh. The fresh carrots shall be U.S. No. 2 or better of the U.S. Standards for Grades of Carrots for Processing (see 6.3).

3.2.3.2 Carrots, diced, individually quick frozen (IQF). The IQF diced carrots shall be prepared from carrots meeting the requirements of 3.2.3.1 and 3.3.3.1. The IQF carrot dices shall be of the latest season's pack. Packaged IQF carrot dices shall be free flowing from the package and shall not show evidence of thawing and refreezing.

3.2.4 Peas, green.

3.2.4.1 Peas, green, fresh, shelled. The fresh shelled green peas shall be U.S. No. 2 or better of the U.S. Standards for Grades of Fresh Shelled Peas for Canning or Freezing. The peas shall be of such size as not to pass through a 9/32 inch sieve.

3.2.4.2 Peas, green, shelled, individually quick frozen (IQF). The IQF shelled green peas shall be prepared from fresh green peas meeting the requirements of 3.2.4.1 and 3.3.4.1. The IQF shelled green peas shall be of the latest season's pack. Packaged IQF peas shall be free flowing from the package and shall not show evidence of thawing and refreezing.

3.2.5 Starch, food, modified, high opacity. The high opacity starch shall be a white, odorless, finely pulverized, modified waxy maize food starch for use in thermostabilized foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140°F to 170°F and be fully hydrated at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The cooked, modified starch slurry shall be bland with essentially no cereal or starch taste.

3.2.6 Tomato paste. The tomato paste shall be U.S. Grade A of the U.S. Standards for Grades of Tomato Paste.

3.2.7 Margarine. The margarine shall be of vegetable origin only and shall conform to the U.S. Standards of Identity for Margarine and shall possess a fine and pleasing flavor. The body shall be smooth, firm, and homogeneous. The margarine shall possess a uniform yellow color and may be salted or unsalted.

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3.2.8 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents and shall comply with purity standards for sodium chloride of the Food Chemicals Codex.

3.2.9 Sugar, white, granulated. The sugar shall be white, refined, granulated cane or beet sugar, or a combination thereof, and shall comply with MIL-STD-900.

3.2.10 Onion powder. The onion powder shall be Fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.11 Vegetable protein, hydrolyzed. Hydrolyzed vegetable protein shall be obtained by the hydrolysis of any one or any combination of corn, soybean, or wheat proteins from which none of the monosodium glutamate has been extracted during processing. The product shall possess a characteristic color and flavor. Salt (chloride calculated as NaCl) shall not exceed 35 percent. The pH of a 2 percent solution shall be not less than 4.3 nor more than 7. Hydrolyzed protein shall be soluble in water at 90°F to the extent of not less than 5 grams in 100 milliliters.

3.2.12 Pepper, black, ground. The ground black pepper shall be ground from the deep brown to black, deep-set, wrinkled, immature berries of Piper nigrum L. The ground pepper shall have a characteristic penetrating odor, a hot biting pungent flavor and a light gray to speckled black-gray color. The ground pepper shall contain not less than 2.0 mL of volatile oil per 100 grams of ground black pepper and be of such size that not less than 95 percent shall pass through a U.S. Standard No. 16 sieve.

3.2.13 Celery, soluble. Soluble celery flavoring shall be prepared by properly blending and dispersing a minimum of 4 percent of total extractives of the dried fruit of Apium graveolens L. and soluble, dry, edible carrier. The flavoring shall possess a clean, celery-like aroma with a slightly bitter taste. The flavoring shall be free flowing, free from caking, and shall be yellowish-gray in color.

3.2.14 Color, caramel. Caramel color shall be a powder or liquid and shall comply with the Food Chemicals Codex.

3.2.15 Garlic powder. The garlic powder shall be Fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.16 Starch, waxy maize, modified (filling and processing aid). The starch shall be a white to off-white, odorless, finely pulverized powder suitable for use as a filling and processing aid in thermostabilized foods. When hyrated at temperatures of 165°F to 200°F, the starch shall develop a paste. The initial viscosity of the paste shall be retained under

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conditions of moderate shear and prolonged holding times. During retorting at temperatures in excess of 200°F, the starch shall undergo extensive to complete breakdown of the initial viscosity. The starch shall be bland and shall demonstrate easy dispersibility.

3.2.17 Sodium tripolyphosphate. Sodium tripolyphosphate shall comply with the Food Chemicals Codex.

3.2.18 Water. Water used for formulation, rehydration, blanching, ice making, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.19 Preblended spice and seasoning mixture. Preblended spices and seasonings may be used. The spices and seasonings in the mixture shall comply with the requirements specified in this specification. The containers used for the spice and seasoning blend shall be labeled with each ingredient and the percentage of each ingredient in the blend. The ingredients shall be in the same proportions as specified in the ingredient formula (see 3.3.6).

3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Beef preparation. Boned and trimmed beef which meets the requirements of 3.2.1 shall be further trimmed, if necessary, to ensure compliance with finished product requirements and shall be processed in accordance with either 3.3.1.1 or 3.3.1.3.

3.3.1.1 Beef processing, log method. The beef shall be mechanically coarse ground through a grinder plate having kidney-shaped openings measuring not less than 2 inches in the shortest dimension. The grinder shall be equipped with a two-blade knife. The beef logs shall be formulated, prepared, and processed as follows:

Ingredients	Percent by weight
Beef	95.75
Ice or ice water	3.00
Salt	1.00
Sodium tripolyphosphate	0.25

a. The coarse ground (chunked) beef shall be mechanically vacuum mixed with the salt and sodium tripolyphosphate. The mixing shall continue until the mixture exhibits a sticky (tacky) consistency. Time from grinding to mixing shall not exceed four hours. Temperature of the meat mixture shall not exceed 40°F during this four hour time period.

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b. The mixed meat shall immediately be tightly stuffed into casings of a type and size to accommodate the cooking method and the finished product meat dimension requirement. Alternatively, the mixed meat may be formed in stainless steel molds to a size which will ensure compliance with finished product meat dimension requirements. The formed beef logs shall be in the cooking process within 24 hours after being formed. The temperature of the logs shall be maintained at an internal temperature of 28°F to 40°F during this 24 hour time period.

c. The beef logs shall be cooked in a cookhouse (smokehouse without smoke) or by other commercially acceptable cooking methods to provide a product meeting the finished product drained weight and other requirements. Water cooking is permitted only for the meat logs in casings provided they are cooked in water until the internal temperature of the log reaches 150°F to 160°F.

d. Immediately after completion of the cooking process, the cooked beef logs shall be placed in an ice water bath or other commercially acceptable rapid cooling method. During cooling the internal temperature of the meat logs shall not remain between 130°F and 80°F for more than 1-1/2 hours, nor between 80°F and 40°F for more than five hours. The cooled logs shall be held for not more than 48 hours at an internal temperature of 28°F to 40°F prior to dicing, or if stored in the frozen form, until packaged and vacuum sealed.

e. The cooked beef may be stored in the frozen log form provided the logs are packaged and vacuum sealed in packaging material complying with the requirements specified in 3.2.1.2. The logs shall be frozen to an internal temperature of 0°F or below within 72 hours of packaging. The packaged, frozen, cooked beef shall be maintained at an internal temperature of 0°F or below for a period not to exceed 60 days prior to pouch filling (see 3.4).

3.3.1.2 Dicing of cooked beef logs. The cooked beef logs shall be mechanically diced to yield uniform dices of 5/8 inch cubes (± 1/16 inch in each dimension). The beef dices shall be screened for elimination of fines. The dices shall be held for no longer than 48 hours at an internal temperature of 28°F to 40°F until pouch filling, or if stored in the frozen form, until packaged and vacuum sealed (see 3.2.1.2). Dices prepared from previously frozen logs shall not be frozen.

3.3.1.3 Beef processing, dicing method. As an alternative to preparation of a log, the dices may be prepared from raw beef. The raw beef shall be diced and processed as follows:

a. The beef shall be diced to yield uniform dices of 5/8 inch cubes (± 1/16 inch in each dimension). The beef dices shall be screened for elimination of fines.

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b. The raw beef dices shall be held for not more than 24 hours at an internal temperature of 28°F to 40°F from time of dicing to the beginning of the cooking process.

c. The beef dices shall be cooked in water, or by other commercially acceptable cooking method, until no raw portions are visible. The cooked beef dices shall be thoroughly drained.

d. The cooked beef dices shall be held for not more than 48 hours at an internal temperature of 28°F to 40°F until pouch filling, or, if stored in the frozen form, until packaged and vacuum sealed (see 3.2.1.2).

3.3.2 Potato preparation.

3.3.2.1 Preparation of fresh potatoes. The fresh potatoes shall be washed, sorted, peeled, and trimmed. The potatoes shall be mechanically diced to yield 3/8 inch cubes (± 1/16 inch in each dimension). The dices shall be blanched sufficiently to prevent discoloration and to remove excess air. The blanched potatoes shall be immediately cooled to the initial temperature of the cooling water and thoroughly drained. The cooled, drained potato dices shall be handled in a manner such as to prevent discoloration and filled into the pouch within four hours after blanching.

3.3.2.2 Preparation of dehydrofrozen potatoes. Diced, dehydrofrozen potatoes shall be prepared in a manner so as to ensure compliance with finished product requirements and shall be handled in a manner so as to prevent discoloration.

3.3.3 Carrot preparation.

3.3.3.1 Fresh carrot preparation. The carrots shall be washed thoroughly to remove soil and foreign material and sorted to remove any carrots that are diseased or otherwise damaged. The sound stock shall be peeled in such a manner so as not to cause degradation of the texture. The peeled carrots shall be trimmed to remove discolored or undesirable portions. The peeled and trimmed carrots shall be mechanically diced to yield 3/8 inch cubes (± 1/16 inch in each dimension). The diced carrots shall be blanched only to the extent necessary to prevent discoloration and to remove excess air. The blanched carrots shall be immediately cooled to the initial temperature of the cooling water and thoroughly drained. The cooled, drained carrots shall be handled in such a manner as to prevent discoloration and filled into the pouches or individually quick frozen within four hours after blanching.

3.3.3.2 Frozen carrot preparation. Frozen carrot dices shall not be blanched, but may be thawed/tempered in such a manner that will not cause further degradation of the texture and only to the extent necessary to accommodate the filling operation. The thawed/tempered carrot dices shall be handled in such a manner as to prevent discoloration.

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3.3.4 Pea preparation.

3.3.4.1 Fresh green pea preparation. The peas shall be washed thoroughly to remove soil and foreign material and sorted to remove any peas that are diseased or otherwise damaged. The peas shall be blanched only to the extent necessary to prevent discoloration and to remove excess air. The blanched peas shall be immediately cooled to the initial temperature of the cooling water and thoroughly drained. The cooled, drained peas shall be handled in such a manner as to prevent discoloration and filled into the pouches or individually quick frozen within four hours after blanching.

3.3.4.2 Frozen pea preparation. Frozen peas shall not be blanched, but may be thawed/tempered in such a manner that will not cause further degradation of the texture and only to the extent necessary to accommodate the filling operation. The thawed/tempered peas shall be handled in such a manner as to prevent discoloration.

3.3.5 Filling and processing aid starch preparation. When a filling and processing aid starch is used, a solution (slurry) shall be prepared with the filling and processing aid starch and cold water in a concentration adequate to hold ingredients in suspension for uniform distribution during mixing and pumping of product.

3.3.6 Sauce preparation. The sauce shall be formulated and prepared as follows:

Ingredients	Percent by weight
Water	79.85
Starch, food, modified, high opacity	5.88
Tomato paste (30 percent solids) 1/	5.88
Margarine	4.41
Salt 2/	1.50
Sugar, white, granulated	0.94
Onion powder	0.59
Vegetable protein, hydrolyzed	0.32
Pepper, black, ground	0.24
Celery, soluble	0.18
Color, caramel 3/	0.18
Garlic powder	0.03

1/ The tomato paste shall be adjusted, as required, to produce a 30 percent tomato soluble solids level.

2/ The total amount of salt in the sauce formula shall be adjusted as necessary to produce a product that complies with the finished product salt requirements (see 3.6).

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3/ The total amount of caramel color shall be adjusted as necessary to produce a product that complies with the finished product color requirement (see 3.6).

NOTE: The following sauce preparation procedures were used in the development of this product. Alternate procedures may be used provided finished product requirements are met. (When alternate procedures are used, the time and temperature requirements specified for the prepared sauce are still applicable.)

a. A slurry shall be made using the starch and part of the water.

b. The remaining ingredients shall be mixed together and heated to not less than 140°F.

c. The starch slurry shall be added and the final mixture heated to a temperature of 150°F to 180°F and held in this temperature range for five minutes.

d. The volume of the final mixture shall be adjusted with water to compensate for evaporation loss during heating and holding.

e. The sauce shall be filled within four hours after cooking. As a final step before filling, a filling and processing aid starch solution (see 3.3.5) may be uniformly blended into the sauce in the amount required to facilitate uniform ingredient distribution in a single-stage filling operation when dehydrofrozen potatoes are used. If a hot fill method is used, the prepared sauce shall be held in the temperature range of 150°F to 180°F at all times prior to pouch filling. If a cold fill method is used, the prepared sauce shall be cooled immediately and held at an internal temperature of 28°F to 40°F at all times prior to pouch filling.

3.3.7 Product preparation. The diced cooked beef, vegetables, and sauce shall be combined in the following approximate proportions using either a single or multistage filling operation. The percent by weight of ingredients may be adjusted if necessary to ensure compliance with finished product drained weight requirements.

Ingredients	Percent by weight
Beef, diced, cooked	40.75
Sauce	33.80
Potatoes, diced 1/	15.00
Carrots, diced	7.50
Peas	2.95

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1/ When dehydrofrozen potatoes are used, the 15 percent by weight shall consist of 7.5 percent by weight of dehydrofrozen potatoes and 7.5 percent by weight of water.

3.4 Pouch filling and sealing. Each pouch (see 5.1) shall be filled with product such as to conform to the finished product requirements and to the following requirements:

a. Each pouch shall be filled and sealed in accordance with the pouch filling and sealing requirements specified for class 1 of MIL-P-44073.

b. A hot fill system is permitted if compatible with the single-stage or multistage filling and sealing method used.

c. The temperature of the diced beef, potatoes, carrots, and sauce at time of filling shall be as specified (see 3.3.1, 3.3.2, 3.3.3, and 3.3.6) for the applicable filling process used.

d. The temperature of the product at time of pouch filling shall be 150°F to 180°F.

e. Each filled and sealed pouch shall be in the retort process within two hours after sealing.

3.5 Pouch thermoprocessing. The filled and sealed pouches shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved. The thermoprocessing operation shall be in compliance with the thermoprocessing requirements for class 1 of MIL-P-44073.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. No individual pouch shall contain less than 7.5 ounces of product.

e. The average net weight shall be not less than 8.0 ounces.

f. Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in any individual pouch shall be not more than 0.35 ounce.

g. No individual pouch shall contain a bone piece measuring 0.3 inch or more in any dimension.

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h. No individual pouch shall contain less than 2.0 ounces drained weight of beef.

i. The average drained weight of beef shall be not less than 2.4 ounces.

j. No individual pouch shall contain less than 1.2 ounces drained weight of vegetables (combined).

k. The average drained weight of vegetables (combined) shall be not less than 1.6 ounces.

l. Of the 2.0 ounces drained weight of beef, not less than 1.5 ounces shall be whole intact beef dices.

m. Texture of the beef dices shall not be dry, rubbery, or mushy.

n. Texture of the vegetables shall not be mushy, hard, fibrous, or tough.

o. The sauce shall be a moderate brown color.

p. Sauce consistency shall be smooth, without lumps, and shall not be excessively thick or thin.

q. The average fat content of the finished product shall not be greater than 8.0 percent and no individual pouch shall have a fat content greater than 10.0 percent.

r. The salt content of the product in any individual pouch shall be not greater than 1.3 percent nor less than 0.5 percent.

s. The product shall show no evidence of excessive heating (materially darkened or scorched).

3.6.1 Palatability. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The beef component and the finished product shall originate and be produced, processed, and stored in plants regularly operating under Meat and Poultry Inspection Regulations of the U.S. Department of Agriculture.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all the requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance of palatability failure shall be cause for rejection of the first article.

4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade Certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods such as the standard methods described in the Official Methods of Analysis of the AOAC and in the Approved Methods of the AACC, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or

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of other valid documents shall be cause for rejection of the involved product.

4.5.3 Filled and sealed pouch inspection. Inspection of filled and sealed pouches shall be in accordance with the quality assurance provisions of MIL-P-44073.

4.5.4 Net weight examination. The net weight of the filled and sealed pouches shall be determined by weighing each sample unit on a suitable scale tared with a representative empty pouch. Any individual net weight of less than 7.5 ounces shall be classified as a minor defect. The lot size shall be expressed in pouches. The sample unit shall be one filled and sealed pouch. The inspection level shall be S-3, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5. Results shall be reported to the nearest 0.1 ounce. In addition, the lot shall be rejected if the sample average net weight is less than 8.0 ounces.

4.5.5 Product and drained weight inspection.

4.5.5.1 Product inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature ($65^{O}F$ to $75^{O}F$) after completion of the thermostabilization operation. The pouches shall be heated in $180^{O}F$ to $190^{O}F$ water for 10 minutes, opened, and inspected. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The sample size shall be as indicated by the sampling plan specified in table I. The contents shall be inspected for the defects indicated in table II.

TABLE I. Double sampling plan for product inspection 1/

Lot size (pouches)	Sample size (pouches)	Cumulative sample	Acceptance number	Rejection number
0 to 500,000	20	--	0	2
	20	40	1	2

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

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TABLE II. Product defects 1/ 2/

Category	Defect
Major	
101	Total weight of cartilage, coarse connective tissue, section of tendons or ligaments, and glandular material, collectively, in any individual pouch is more than 0.35 ounce
102	Presence of a bone piece measuring 0.3 inch or more in any dimension
103	Texture of beef dices is dry, rubbery, or mushy
104	Texture of vegetables is mushy, hard, fibrous, or tough
105	Sauce not a moderate brown color
106	Sauce consistency lumpy or is excessively thick or thin
107	Product shows evidence of excessive heating (materially darkened or scorched)

1/ The presence of any foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability and overall appearance shall be cause for rejection of the lot (see 3.6.1).

4.5.5.2 Drained weight inspection. The filled and sealed sample pouches shall be held for a minimum of 72 hours at room temperature (65°F to 75°F) after completion of the thermostabilizing operation. The pouches shall be heated in 180°F to 190°F water for approximately 10 minutes, opened, and inspected. The lot size shall be expressed in pouches. The sample unit shall be the contents of one pouch. The sample size shall be as indicated by the double sampling plan specified in table III. The drained weight shall be determined and inspected for the defects listed in table IV.

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TABLE III. Double sampling plan for drained weight inspection 1/

Lot size (pouches)	Sample size (pouches)	Cumulative samples	Acceptance number	Rejection number
0 to 500,000	20	--	0	2
	20	40	1	2

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE IV. Drained weight defects 1/

Category	Defect
Major	
101	Drained weight of beef in a pouch is less than 2.0 ounces 2/
102	Drained weight of vegetables (combined) in a pouch is less than 1.2 ounces 3/
103	Of the 2.0 ounces drained weight of beef, less than 1.5 ounces are whole intact beef dices 4/

1/ To determine the drained weight, the pouch contents shall be poured into a flat bottom container. A minimum of three times the pouch's volume of 180°F to 190°F water shall be added to the container so as to cover the contents. The contents and water shall be gently agitated such as to liquify rendered fat and to remove the sauce without breaking the beef dices or vegetables. The contents shall then be poured into a U.S. Standard No. 7 sieve in a manner that will distribute the product over the sieve without breaking the beef dices or vegetables. The sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted at an approximate

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45° angle and allowed to drain for two minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight of the beef and vegetables (combined) shall be determined individually and shall be reported to the nearest 0.1 ounce.

2/ If the sample average drained weight of beef is less than 2.4 ounces, the lot shall be rejected.

3/ If the sample average drained weight of vegetables (combined) is less than 1.6 ounces, the lot shall be rejected.

4/ For purposes of this examination, the definition of a whole intact beef dice is one that measures not less than 1/4 inch in each of two separate dimensions.

4.5.6 Fat and salt content testing. Eight filled and sealed pouches shall be selected at random from the lot. The product shall be tested for fat content in accordance with the Official Methods of Analysis of AOAC, method 976.21, 960.39, or 985.15, and for salt content in accordance with the Official Methods of Analysis of AOAC, method 935.47, except that preparation of the samples shall be as follows: The unopened pouches shall be gently warmed in a water bath to melt fat adhering to the inside of the pouches. The pouches shall be opened and the entire contents of each pouch shall be separately blended in a Waring Blendor or equivalent. The test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the fat or salt requirements in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

4.5.7 Packaging examination. Examination of packaging shall be in accordance with the quality assurance provisions of MIL-P-44073.

5. PACKAGING

5.1 Packaging. Packaging shall be in accordance with the requirements of MIL-P-44073.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DOSISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

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c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.3 Fresh carrots for further processing. It has been shown that carrot varieties which have a low density such as Imperator or Red Core Chantenay perform satisfactorily in this product.

6.4 Subject term (key word) listing.

Combat field feeding
Food processing
Operational ration
Pouch food
Shelf stable

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project 8940-0744

Review activities:

Army - MD, QM
Navy - MC
DP - SS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-44059D	1993 September 3

3. DOCUMENT TITLE

BEEF STEW, THERMOSTABILIZED, FOR MEAL, READY-TO-EAT

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

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U.S. Army Natick RD&E Center	(1) Commercial 508-651-4501	(2) AUTOVON/DSN 256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WT Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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